

116 Advanced Statistics for Neuroscience

Exercise 02

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Red blood cell counts

Use Matlab or Python or something similar to test the null hypothesis that the **red blood cell** counts in two groups of mice are statistically identical. Your cell counts were automated and came in units of millions of cells per microliter of serum.

Your group A comprised $n_A = 12$ animals with the following counts

$$\mathbf{x}_A = \{14.9, 5.6, 5.1, 1.0, 11.6, 11.5, 7.7, 11.6, 8.7, 9.2, 8.5, 9.8\}$$

Your group B included $n_B = 14$ animals with the following counts

$$\mathbf{x}_B = \{12.5, 25.9, 6.2, 2.7, 6.3, 6.5, 9.8, 11.2, 10.9, 14.7, 13.9, \\ 15.8, 20.9, 18.1\}$$

Means, variances, and t-statistic

Compute the mean and the variance for the RBC in each group. You need to use Bessel's correction for the variance, because you have only the sample mean, not the true mean.

Compute the t-statistic for an independent (unpaired) t-test and obtain the associated p-value (using `tcdf` in Matlab or `t.cdf` in Python). The p-value is the **area of the tail**, starting with your observed t-statistic!

So what do you conclude? Can you rule the null hypothesis out?

Visualization

Plot the t-distribution for $n_A + n_B - 2$ degrees of freedom. Note that you 'lose' two degrees of freedom by using the sample mean for each group.

For comparison, add the standard normal distribution to the same plot. The difference between the two distributions will not be large!

Plot the observed t-statistic along the x-axis. Does it fall near the middle or in one of the tails?

Finally, along the x-axis, add the 'significance' thresholds for the left and right tails, $p = 0.025$ and $p = 0.975$. You will have to get the x-position of these thresholds from the inverse cumulative distribution (`tinverse` or `t.ppf`).